

Proof. We get this by applying Proposition 6.8 both to $\text{Dep}(V)$ and to $\text{Val}(V)$, which are dual to each other by Proposition 6.4. $\square$

Therefore two (labeled) configurations of vectors or points have *the same oriented matroid* if they have the same set of circuits, or (equivalently) the same set of cocircuits, the same set of vectors, or the same set of covectors.

(c) Duality

A concept of *duality* is built into the whole structure of oriented matroids. In fact, since the vectors and the covectors of an oriented matroid $\mathcal{M}(V)$ arise in the same way as the sign vectors of a subspace, they also satisfy the same axioms (ignoring a switch from row vectors to column vectors).

Definition 6.10. The *dual* of an oriented matroid $\mathcal{M}$ is the oriented matroid $\mathcal{M}^*$ with the following properties:

- The vectors of $\mathcal{M}^*$ are the covectors of $\mathcal{M}$, and thus the circuits of $\mathcal{M}^*$ are the cocircuits of $\mathcal{M}$.
- The covectors of $\mathcal{M}^*$ are the vectors of $\mathcal{M}$, and thus the cocircuits of $\mathcal{M}^*$ are the circuits of $\mathcal{M}$.

From the way we have defined (realizable) oriented matroids, it is clear that for every oriented matroid $\mathcal{M}$, there is a unique dual oriented matroid $\mathcal{M}^*$, whose dual is

$$(\mathcal{M}^*)^* = \mathcal{M}.$$

In fact, $\mathcal{M} = \mathcal{M}(U)$ is realizable with $U \subseteq \mathbb{R}^n$, then $\mathcal{M}^* = \mathcal{M}(U^\perp)$, for the orthogonal space

$$U^\perp := \{\mathbf{c} \in (\mathbb{R}^n)^* : \mathbf{c}\mathbf{x} = 0 \text{ for all } \mathbf{x} \in U\}.$$

So existence and uniqueness of the dual follow from Proposition 6.4 (in the realizable case):

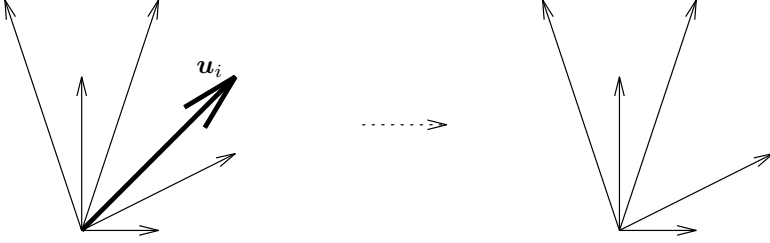
$$\mathcal{M}^* = \mathcal{M}(\text{Dep}(V)) \quad \text{for} \quad \mathcal{M} := \mathcal{M}(\text{Val}(V)).$$

Furthermore, if $\mathcal{M}$ has rank r , then $\mathcal{M}^*$ has rank $n - r$, and conversely.

(d) Deletion and Contraction

There are two very natural and fundamental operations on point configurations: *deletion* and *contraction*. They are dual to each other, and they directly translate into oriented matroid language and terminology.

Consider a vector configuration $V \in \mathbb{R}^{r \times n}$, and let $\mathbf{u}_i \in V$. We can certainly delete $\mathbf{u}_i$ from V , to get the new configuration $V \setminus \mathbf{u}_i$.

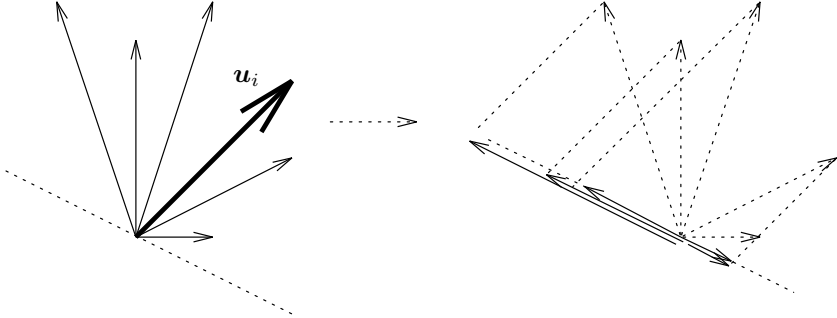


We get value vectors on $V \setminus \mathbf{u}_i$ from those on V by simply deleting the components corresponding to $\mathbf{u}_i$, while dependences and circuits of $V \setminus \mathbf{u}_i$ are those dependences, respectively circuits, of V which do not involve $\mathbf{u}_i$ (i.e., have zero i -component). This proves the following result, where we label the vectors in V (the “ground set” of $\mathcal{M}(V)$) by $\{1, \dots, n\}$, so we write $\mathcal{M} \setminus i$ for $\mathcal{M}(V \setminus \mathbf{u}_i)$.

Proposition 6.11. *The oriented matroid $\mathcal{M}(V) \setminus i$ of $V \setminus \mathbf{u}_i$ is given as follows:*

$$\begin{aligned} \mathcal{V}(V \setminus \mathbf{u}_i) &= \{v \setminus i : v \in \mathcal{V}(V), v_i = 0\} & \mathcal{V}^*(V \setminus \mathbf{u}_i) &= \{v \setminus i : v \in \mathcal{V}^*(V)\} \\ \mathcal{C}(V \setminus \mathbf{u}_i) &= \{c \setminus i : c \in \mathcal{C}(V), c_i = 0\} & \mathcal{C}^*(V \setminus \mathbf{u}_i) &= \text{MIN}\{c \setminus i : c \in \mathcal{C}^*(V)\}. \end{aligned}$$

The dual operation is the contraction of $\mathbf{u}_i$: for this we project V parallel to $\mathbf{u}_i$ to some hyperplane that does not contain $\mathbf{u}_i$. If $\mathbf{u}_i = \mathbf{0}$, then we just delete $\mathbf{u}_i$.



Algebraically, we can do this by choosing a linear function $\mathbf{c} \in (\mathbb{R}^r)^*$ such that $\mathbf{c}\mathbf{u}_i \neq 0$ (for example, $\mathbf{c} := \mathbf{u}_i^t$ will do), and map

$$\mathbf{u}_j \longmapsto \bar{\mathbf{u}}_j := \mathbf{u}_j - \frac{\mathbf{c}\mathbf{u}_j}{\mathbf{c}\mathbf{u}_i} \mathbf{u}_i.$$

This yields the new configuration

$$V/\mathbf{u}_i := \{\bar{\mathbf{u}}_1, \dots, \bar{\mathbf{u}}_{i-1}, \bar{\mathbf{u}}_{i+1}, \dots, \bar{\mathbf{u}}_n\}$$

in the hyperplane $\{\mathbf{v} \in \mathbb{R}^r : \mathbf{c}\mathbf{v} = 0\}$.

We get dependences on $V/\mathbf{u}_i$ from those on V by simply deleting the components corresponding to $\mathbf{u}_i$, while value vectors of $V/\mathbf{u}_i$ are those value vectors V which are zero on $\mathbf{u}_i$ (i.e., have zero i -component).

Proposition 6.12. *The oriented matroid $\mathcal{M}(V)/i$ of $V/\mathbf{u}_i$ is given as follows:*

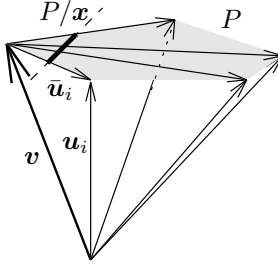
$$\begin{aligned} \mathcal{V}(V/\mathbf{u}_i) &= \{\mathbf{v} \setminus i : \mathbf{v} \in \mathcal{V}(V)\} & \mathcal{V}^*(V/\mathbf{u}_i) &= \{\mathbf{v} \setminus i : \mathbf{v} \in \mathcal{V}^*(V), v_i = 0\} \\ \mathcal{C}(V/\mathbf{u}_i) &= \text{MIN}\{\mathbf{c} \setminus i : \mathbf{c} \in \mathcal{C}(V)\} & \mathcal{C}^*(V/\mathbf{u}_i) &= \{\mathbf{c} \setminus i : \mathbf{c} \in \mathcal{C}^*(V), c_i = 0\}. \end{aligned}$$

Let us mention two examples that show how deletion and contraction appear in connection with polytopes.

Examples 6.13. Let $P \subseteq \mathbb{R}^d$ be a polytope, let $X := \text{vert}(P)$ be its vertex set, and let $V \in \mathbb{R}^{r \times n}$ be the corresponding vector configuration in $\mathbb{R}^r$ ($r = d+1$).

If F is a face of P , then the vector configuration for F is obtained by deleting from V all the vertices that do not lie on F .

If $\mathbf{x} \in \text{vert}(P) \subseteq \mathbb{R}^d$ is a vertex of P and $\mathbf{v} \in V \subseteq \mathbb{R}^r$ is the corresponding vector, then the vector configuration of the vertex figure $P/\mathbf{x}$ is the contraction $V/\mathbf{v}$. (In this case the projection hyperplane for the contraction can be taken parallel to the hyperplane spanned by P .)



Note that by contracting we get a vector configuration that may also represent a lot of interior points of the vertex figure, corresponding to vertices $\mathbf{w} \in \text{vert}(P)$ such that $[\mathbf{w}, \mathbf{v}]$ is not an edge of P .

6.4 Dual Configurations and Gale Diagrams

Now observe that $\text{Dep}(V) \subseteq \mathbb{R}^n$ determines the configuration $V \in \mathbb{R}^{r \times n}$ of column vectors uniquely up to coordinate transformations in $\mathbb{R}^r$, which correspond to row operations on the matrix V . Thus the dual space $\text{Val}(V)$ determines a configuration of row vectors in $(\mathbb{R}^{n-r})^*$, which completely encodes the vector configuration V .